GARCH Audit

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Volatility asymmetry of the S&P 500, weekly returns

We audit the constant-mean GARCH(1,1) written by the AI on weekly S&P 500 log-returns in percent and compare it with the Python `arch` package. We then study volatility asymmetry with the GJR-GARCH and the news impact curves. *Labels*: [c] computed (repository code, sheet `weekly_spx_ust10`), [a] assumed; no number is simulated or taken from an AI answer. *Slide n*: course deck (NLNPE); *handout p. n*: tutorial handout.

1. Audit

Table 1: the six checkpoints (handout p. 11), on the AI's initial code

Checkpoint	Verdict	Reference	Verification
Residuals	Correct	slide 34	The recursion uses $e_t = r_t - \mu$, not r_t.
Initial variance	Correct	slide 36	Stated: sample variance, divisor T [a], $h_1 = 5.2774$ [c]. The course alternative is $\omega/(1-\alpha-\beta)$.
Likelihood	Correct	slide 39	Minimises $-\sum_t \log \ell_t$ (Gaussian): correct sign.
Constraints	Correct	slide 36	Exponential/softmax reparametrisation: $\omega > 0$, $\alpha, \beta > 0$, $\alpha + \beta < 1$, no clipping. Exact zeros are excluded.
Standard errors	Correct	slides 40-41	Numerical Hessian: a valid estimator of $I(\theta)$. Recomputed at the optimum, it gives plausible t (7.3 to 28.4 [c]) and matches `arch`'s classic covariance to 10^{-4} [c]. BHHH (slide 46) kept as a cross-check.
Scale	Correct	slide 56	The code expects returns in percent, as the prompt asks; we feed it $r_t = 100 \log(P_t/P_{t-1})$, hence $\hat{\omega} = 0.32$ [c].

The rest of the report uses the final version (BHHH standard errors, Hessian as a cross-check).

Table 2: GARCH(1,1) estimates [c], our implementation and the `arch` package

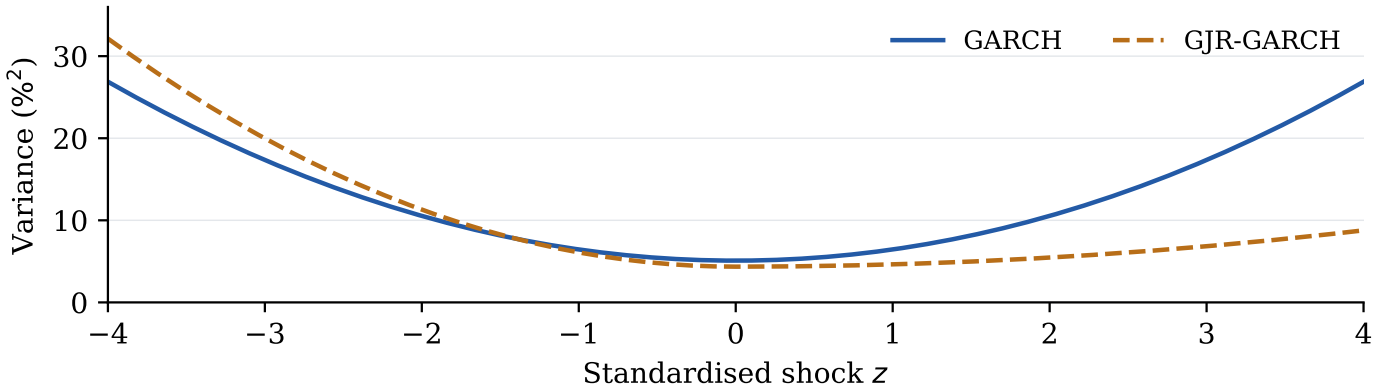
	Our implementation	BHHH s.e.	`arch` GARCH	Absolute gap	Relative gap (%)
μ	0.260789	0.035543	0.260804	0.000015	0.01
ω	0.320176	0.048116	0.320190	0.000014	0.00
α	0.211234	0.015736	0.211268	0.000034	0.02
β	0.739120	0.018465	0.739098	0.000022	0.00

The coefficients agree to two decimals. The log-likelihoods are -5084.787 [c] and -5084.785 [c]. For ω, the BHHH (0.048 [c]), Hessian (0.063 [c]) and robust (0.085 [c]) standard errors diverge: fat tails (excess kurtosis 6.4 [c]) call for caution with Gaussian inference. ARCH-LM rejects a constant variance (12 lags [a], $LM = 374.9$ [c], $p = 7.6 \times 10^{-73}$ [c]), then finds no ARCH effect left after filtering ($p = 0.712$ [c]).

2. Asymmetry

GJR estimated with `arch` (slide 52), robust standard errors: $\hat{\mu} = 0.172$ [c], $\hat{\omega} = 0.368$ [c], $\hat{\alpha} = 0.052$ [c], $\hat{\beta} = 0.744$ [c] and $\hat{\gamma} = 0.272355$ [c] (s.e. 0.056559 [c], $t = 4.815$ [c], $p = 1.47 \times 10^{-6}$ [c], asymptotic). The correlation between this week's return and the absolute value of next week's return is -0.2243 [c] (2,386 pairs): falls tend to precede larger moves, without establishing causality.

We hold the current variance at its unconditional level (handout p. 15), the $\gamma/2$ assuming symmetric innovations [a]: $\bar{h}_G = \omega/(1-\alpha-\beta) = 6.449$ [c] and $\bar{h}_J = \omega/(1-\alpha-\beta-\gamma/2) = 5.360$ [c] ($\%^2$).



Checks of handout p. 15: (1) ✓ curves drawn at $\bar{h}$, not at the last h_t; (2) ✓ indicator on $z < 0$ and $\gamma/2$ in $\bar{h}_J$; (3) × the curves do not coincide for $z > 0$, yet the asymmetry is not on the wrong side: the GARCH spreads over both sides a response the GJR keeps for negative shocks ($\hat{\alpha}_G = 0.211$ vs $\hat{\alpha}_J = 0.052$). Next-week variance at $z = +2$: 10.5 [c] (GARCH) vs 5.5 [c] (GJR); at $z = -2$: 10.5 [c] vs 11.3 [c].

Conclusion. The AI got one point wrong that the course settles: pressed on its numerical-Hessian standard errors, it called them wrong, whereas the Hessian estimates the information matrix just as BHHH does (slides 40-41 and 46).